

SUMMARY TABLE: MATHEMATICS GRADE 10

Sub-Strand 4.1: Statistics I — Teacher Reference (Pre-filled)

SUMMARY TABLE: MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 4.1: Statistics I
Driving Question	DRIVING QUESTION: How can you use the two sets of data to predict the learners' and the teachers' next steps?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Two Sets of Scores: Phenomenon Launch	We observed that the same 20 learners scored between 67 and 95 in Subject 1 (range 28, all 20 pass) and between 11 and 60 in Subject 2 (range 49, only 4 pass and 16 fail at a pass mark of 50). Ordering the scores made the highest, lowest and pass/fail easy to see.	We learned that arranging raw data in ascending order shows the highest, lowest and pass/fail at a glance, but it does not clearly show patterns, spread or the typical score. Reasons for the difference are only hypotheses until we have more data.	The difference is between the subjects, so we cannot say all 20 learners are weak. To decide next steps for the learners, Mr. Otieno and Mrs. Wanjiku, we need to organise and summarise the data with tables, typical values and graphs, which is what we will do in the coming lessons.
Lesson 2 Frequency Distribution Tables for Ungrouped Data: Tallying the Two Sets of Scores	We tallied the 20 scores in Subject 1 and the 20 scores in Subject 2 into frequency tables with columns Score, Tally and Frequency. In Subject 1 every score occurred once (20 different scores). In Subject 2, 32 and 60 each occurred twice and the other scores once (18 different scores). Both frequency columns added to 20.	A frequency distribution table for ungrouped data lists each score, a tally (grouped in fives) and the frequency, and the frequencies must add up to $n = 20$. Tables help us count accurately and spot repeats, but ungrouped tables with many different scores are long and do not yet clearly show where scores cluster or how widely they spread.	The tables are accurate summaries of how often each score occurs, which helps Madam Achieng, Mr. Otieno and Mrs. Wanjiku check the data. However, Subject 1's table is as long as its list, so we need another way of organising the scores to see patterns and spread. The tables cannot tell us why the subjects differ; that would need more data.
Lesson 3 Grouped Frequency Distribution Tables: Choosing Class Intervals and Classifying the Scores	We organised the 20 scores of Subject 1 and Subject 2 into grouped frequency tables using a common class width of 10 (classes 10-19 to 90-99). Subject 1 frequencies: 0, 0, 0, 0, 2, 7, 8, 3 and Subject 2 frequencies: 4, 4, 5, 3, 2, 2, 0, 0, 0, each totalling 20.	We learned how to use the range (28 and 49) to choose a class width, write class limits, find class boundaries (9.5, 19.5, ..., 99.5) and midpoints (14.5, ..., 94.5), and classify scores with tally marks so that frequencies add to $n = 20$.	Grouping shows that Subject 1 scores cluster high (70-89 has 15 of 20 learners and no score is below 60), while Subject 2 scores are spread out (16 of 20 are below 50). The raw lists hid this pattern. The tables do not tell us why, so we need more data, such as topic-by-topic marks and a second test, before deciding the causes and next steps.

Lesson 4 Mean, Mode and Median of Ungrouped Data: Comparing the Two Subjects	Calculating from the ordered lists and frequency tables, we observed that Subject 1 has a sum of 1607, mean 80.35, median 80.5 and no mode, while Subject 2 has a sum of 677, mean 33.85, median 31.5 and two modes (32 and 60).	We learned how to calculate the mean (sum of x divided by n), the median (average of the 10th and 11th ordered values when $n = 20$) and the mode (the most frequent score, which may be none or more than one), and that each gives a different way to describe a typical score.	The typical score in Subject 1 (about 80) is well above the pass mark of 50, while the typical score in Subject 2 (about 32-34) is below it, so the difference seems to be between subjects and not because the learners are weak overall; the measures may help predict next steps but they do not show spread or the cause, so we need more data.
Lesson 5 Mean, Mode and Median of Grouped Data: Estimating the Centre of Each Subject	We observed that the grouped tables let us estimate the mean (Subject 1: $1610/20 = 80.5$; Subject 2: $700/20 = 35$), the modal class and mode (80-89, about 81.2; 30-39, about 32.8) and the median (about 80.75 and 33.5). These are close to the exact values from Lesson 4 but not identical.	We learned how to use midpoints and fx to estimate the mean, boundaries and $d1$, $d2$ to estimate the mode, and cumulative frequency to estimate the median. We learned that grouped measures are estimates because each score is replaced by its class midpoint or position in the class.	We can explain that Subject 1 is centred around 80 and Subject 2 around 34, so decisions must be made per subject: the same learners cannot all be 'weak'. We cannot yet say why the scores differ, and we need spread information and more data before deciding the full next steps.
Lesson 6 Representation of Data I: Histograms and Frequency Polygons	We drew histograms (bars touching at the boundaries 9.5 to 99.5) and frequency polygons (points at midpoints 14.5 to 94.5, closed to the axis) for Subject 1 and Subject 2 on the same axes. Subject 1: scores clustered between 60 and 99, tallest bar 80-89 ($f = 8$). Subject 2: scores spread from 10 to 69, tallest bar 30-39 ($f = 5$).	A histogram shows the frequency of each class with bars at the class boundaries; a frequency polygon joins (midpoint, frequency) points and is closed at the empty classes either side. Using the same classes and scales for both subjects makes the comparison fair, and the graphs make shape and spread easy to see.	The graphs show that Subject 1 is high and tightly clustered while Subject 2 is low and widely spread, so the difference is between subjects and not about the learners all being weak. The graphs cannot tell us why the scores differ, so we need more data (for example topic-by-topic marks and a second test) before concluding.
Lesson 7 Representation of Data II: Cumulative Frequency and Ogives	We observed that the cumulative frequency of Subject 2 reaches 13 by 39.5 and 20 by 69.5, while Subject 1 stays at 0 until 59.5 and then rises to 20 by 99.5. Both ogives were drawn using (upper boundary, cf) points. From the ogives we read estimates: Subject 1 Q1 73.8, median 80.75, Q3 87.0; Subject 2 Q1 22.0, median 33.5, Q3 46.2.	We learned that a cumulative frequency table adds up the frequencies to show how many learners scored below each upper boundary, and that an ogive lets us read the median (10th), Q1 (5th) and Q3 (15th) as estimates. The IQR = $Q3 - Q1$ is 13.2 for Subject 1 and 24.2 for Subject 2, so the middle half of Subject 2 scores is more spread out.	The ogives help us to answer the driving question by showing position and spread: Subject 1 scores are high and close together, while Subject 2 scores are lower and more spread out, with a median estimate of 33.5. This supports predictions that Mr. Otieno's class may continue with extension work and that Mrs. Wanjiku and the learners may reteach and revise, give support to the learners below 50 and collect more data. The graphs show what happened but not why, so we need more data such as topic-by-topic marks and a second test before concluding.
Lesson 8 Interpretation of Data: Comparing Subjects and Predicting Next Steps	We observed that Subject 1 scores are high and tightly clustered (mean 80.35, median 80.5, range 28, all 20 learners pass) while Subject 2 scores are low and widely spread (mean 33.85, median 31.5, range 49, only 4 of 20 pass, 16 below 50, 7 at 40 or more).	We learned that tables, measures and graphs must be read together to compare two sets of data using average, spread, shape and pass rate. We learned that the grouped values are estimates, and that the data shows differences between subjects	We explain that the same 20 learners cannot all be weak, because they all passed Subject 1 but most failed Subject 2, so decisions must be made per subject. We predict that Mr. Otieno's class should continue with extension or enrichment work,

	The histograms, polygons and ogives show the same pattern: the Subject 1 graphs are narrow and to the right, and the Subject 2 graphs are wide and to the left.	but cannot tell us why the scores differ.	and that Mrs. Wanjiku and the learners should reteach and revise the Subject 2 topics, give remedial support to the 16 learners below 50, and collect more data (such as topic-by-topic marks and a second test) before concluding why the scores differ.
Lesson 9 Why Collecting, Organising and Interpreting Data Leads to Effective Decisions	Same 20 learners: Subject 1 scores 67 to 95 (mean 80.35, median 80.5, range 28, all 20 pass); Subject 2 scores 11 to 60 (mean 33.85, median 31.5, range 49, only 4 pass, 16 below 50, 7 at 40 or above). Tables, measures and graphs showed these patterns clearly, while the raw lists hid them.	Collecting, organising (ordering, frequency tables) and interpreting (measures and graphs) data shows centre, spread and pass rates, so decisions can be based on evidence. Because the same learners did well in one subject and poorly in the other, decisions must be made per subject. The data cannot tell us why, so causes are hypotheses to be investigated.	Mr. Otieno's class should continue with extension or enrichment work; Mrs. Wanjiku and the learners should reteach and revise the Subject 2 topics and run remedial support for the 16 learners below 50; and more data (topic-by-topic marks, a second test) should be collected to find the cause before concluding. Organised and interpreted data lets us predict and justify next steps for learners and teachers.

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